
(a) Ask the database first — BIRD Q650, original data
1
Cheap model, 3.3 s
Hint: "Outliers is the name of the badge"
SELECT LastAccessDate, ...
FROM users JOIN badges
WHERE Name = 'Outliers'
2
Database evidence, 54 ms
bounded exec
0 rows
value probe
not found
near value
'outliers'
the SQL text looked fine; the data says otherwise
3
Plan: value − λ · cost
value fix
+0.16
resample
-0.08
repair(E)
-0.15
switch pro
-0.63
V(s0) = 0.033; the DB-only fix (cost 0.01) beats every LLM call
4
Re-plan and select
K1: 0 rows
c0 (cheap) strong model repeats 'Outliers'
P(K2 ≻ K1) = 0.994
K2: 1 row
value fix 'outliers' ✓ correct
1
LLM call
2.50
cost units
3.4 s
end-to-end
0.31
V(s1) after fix
every LLM-only plan repeats the wrong literal
(b) Know when not to ask — Q368, 100× data
1
"What is the percentage of borderless cards?"
SELECT COUNT(CASE WHEN bordercolor='borderless'
THEN id END)*100.0/COUNT(id) FROM cards
2
Price it, then decide
EXPLAIN cost 8.3e+05
bounded exec
skip
pred. 2.96 s · VOI -0.004
value probe
skip
pred. 0.18 s · VOI -0.006
no outcome could change the decision
3
DB time, same answer ✓
fixed bundle
7.1 s
optim.-gated
7.1 s
adaptive
1 ms
Accept immediately: 7.1 s of database time saved.
On 100× data, at 1.25–1.5× budgets, this cuts mean database time by 82–89%
at equal accuracy (498 held-out questions).